JAMES HARRISON

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PROFESSIONAL SUMMARY

I'm a data scientist with 12 years of hands-on experience building machine learning systems that solve real business problems. I've spent the last decade working across finance, healthcare, and retail sectors, developing everything from predictive models to recommendation engines. I'm most interested in the intersection of data engineering and applied ML, and I'm comfortable diving into poorly specified problems and finding the signal in noisy datasets.

WORK EXPERIENCE

Senior Data Scientist

DataVault Solutions

www.datavault.io

Liverpool, United Kingdom

August 2021 - Present

Designed and deployed a customer churn prediction model that identified at-risk accounts three months in advance, resulting in retention of GBP 2.3m annual revenue through targeted interventions.

Led a team of four junior data scientists and one ML engineer on a multi-year project to build an automated anomaly detection system for transactional data across 15 million daily records.

Established best practices for model evaluation and monitoring, implementing automated retraining pipelines that reduced model drift by 67% and eliminated manual intervention for 80% of production models.

Mentored six data scientists on statistical rigor and experimental design, reducing the rate of false positive findings by 45% across the organization.

Principal Data Scientist

Meridian Healthcare Analytics

www.meridian-health.co.uk

Manchester, United Kingdom

March 2018 - July 2021

Built a machine learning framework for predicting patient readmission risk within 30 days of discharge, achieving 0.82 AUC and helping the NHS trust reduce readmissions by 12%.

Implemented feature engineering pipelines using PySpark to process 500GB of historical patient records, reducing data preparation time from six weeks to three days.

Conducted statistical analysis of treatment outcomes across 40 different medical procedures, working closely with clinical staff to ensure medical validity and actionable insights.

Established a model registry and version control system, enabling reproducibility across 25+ models in production and reducing debugging time during model failures by 60%.

Data Scientist

Retail Analytics Ltd

www.retailanalytics.net

Liverpool, United Kingdom

September 2011 - February 2018

Developed a dynamic pricing algorithm that optimized margins across 300,000 SKUs, increasing gross profit by GBP 4.7m annually while maintaining competitive market positioning.

Created customer segmentation model using RFM analysis and clustering techniques, enabling targeted marketing campaigns with 23% higher conversion rates than untargeted campaigns.

Built inventory demand forecasting models using time series analysis and gradient boosting, reducing excess inventory by 18% and stock-outs by 31%.

Designed A/B testing framework for evaluating product recommendations, running 50+ concurrent experiments and establishing statistical guidelines that prevented premature conclusions.

EDUCATION

PhD in Computer Science

University of Manchester

Manchester, United Kingdom

2008 - 2011

Thesis: Scalable Machine Learning Algorithms for High-Dimensional Data

MSc in Statistics

University of Liverpool

Liverpool, United Kingdom

2006 - 2008

BSc in Mathematics

University of Edinburgh

Edinburgh, United Kingdom

2003 - 2006

TECHNICAL SKILLS

Programming Languages: Python, SQL, Scala, R, Bash

Machine Learning: Regression, Classification, Time Series Analysis, Clustering, Anomaly Detection, Deep Learning, Gradient Boosting, Neural Networks

Data Processing: Apache Spark, Pandas, NumPy, Scikit-learn, TensorFlow, PyTorch

Databases: PostgreSQL, MySQL, Snowflake, Redshift, MongoDB

Tools and Platforms: Jupyter Notebook, Git, Docker, Kubernetes, Jenkins, Airflow, MLflow

Statistics: Hypothesis Testing, A/B Testing, Bayesian Methods, Experimental Design

Data Visualization: Matplotlib, Seaborn, Plotly, Tableau

Cloud Platforms: AWS (SageMaker, EC2, S3), Google Cloud Platform, Azure ML